

PRISMA-ScR Checklist

Reporting standard: PRISMA Extension for Scoping Reviews (PRISMA-ScR) — Tricco AC, Lillie E, Zarin W, et al. PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation. *Ann Intern Med.* 2018;169(7):467-473.

IMPORTANT — verify before submission: Item wording and numbering below were reconstructed from memory of the published checklist, not copied from the primary source. Cross-check every item's exact wording and section grouping against the original Tricco et al. 2018 checklist (freely available at equator-network.org and as the article's supplement) before finalizing — reviewers and editors check submissions against the official checklist PDF directly, and any drift in item wording is an easy, avoidable desk-reject flag.

How to use this file: Each row states (a) whether/where the item is addressed in this project's current state, and (b) a `Page #` placeholder to fill in once the manuscript is typeset. As of 2026-07-26, extraction, synthesis, and manuscript drafting (Abstract through Conclusion) are complete; all 22 items are reported. `Page #` columns remain placeholders pending final typesetting for a specific journal template.

Title

#	Item	Reported?	Location / Notes	Page #
1	Identify the report as a scoping review.	Yes	Title includes "A Scoping Review": <i>"Beyond Fidelity: Evaluating the Validity of Explainable AI in Emergency Department Triage and Disposition — A Scoping Review."</i>	—

Abstract

#	Item	Reported?	Location / Notes	Page #
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2	Structured summary (background, objectives, eligibility criteria, sources of evidence, charting methods, results, conclusions).	Yes	Abstract (`manuscript_v1.md`), structured as Background/Objective/Methods/Results/Conclusions, reporting final numbers (7 included studies) and RQ1-RQ3 findings.	—
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Introduction

#	Item	Reported?	Location / Notes	Page #
3	Rationale — describe the rationale in the context of what is already known; explain why the questions/objectives lend themselves to a scoping (not systematic) review approach.	Yes	Introduction (`manuscript_v1.md`), opening paragraphs plus "Where this review sits relative to existing literature," establishing the interpretability/trust-calibration/evaluation-validity framing, the regulatory-timeliness rationale, and the literature gap this scoping (not systematic) review addresses.	—
4	Objectives — explicit statement of questions/objectives with key elements (population, concepts, context).	Yes	Introduction, "Research questions" (`manuscript_v1.md`): RQ1-RQ3 stated verbatim, with population (EM decision points), concepts (interpretability, trust calibration, evaluation validity), and context (regulatory-evidence expectations) specified.	—

Methods

#	Item	Reported ?	Location / Notes	Page #
5	Protocol and registration — state whether a protocol exists, where accessible, and registration number if available.	Yes (content ready)	OSF registration: `docs/osf/preregistration_draft.md`, filed 2026-05-24, revised to Draft v2.2 (2026-06-22) following the 2026-06-10 EM-scope pivot. PROSPERO explicitly not used — deprioritized 2026-06-10 (`docs/osf/prospero_draft.md` status note; `memos/decision_log.md`), state as a disclosed methodological choice, not an omission, and reframe the rationale for an academic audience rather than citing the grading rubric directly.	—
6	Eligibility criteria — characteristics used as inclusion/exclusion criteria (e.g., years, language, publication status), with rationale.	Yes (content ready)	3-gate boundary in `docs/protocol/inclusion_boundary.md` v2.2: (a) decision point occurs during the initial ED encounter — intake, acuity/ESI scoring, or immediate disposition; (b) an explainable/interpretable AI method is applied; (c) an empirical evaluation component is reported. Date range 2015-01-01 through search-execution date (rolling; extended by the 2026-07-14 differential update through 2026-07-14). English language only. Rationale for the EM-only scope: 2026-06-10 pivot decision (8-week single-reviewer timeline; see decision log).	—
7	Information sources — all sources searched, including dates of coverage and most recent search date.	Yes (content ready)	PubMed (searched 2015-01-01 through 2026-06-10, formally executed; differential update 2026-06-11 through 2026-07-14), IEEE Xplore (executed 2026-06-18), ACM Digital Library (executed 2026-06-22, via OpenAlex API substitution — disclose the methodology deviation and why). Embase considered but dropped (2026-06-22, no institutional access — disclose as limitation). CINAHL considered, shelved per the EM-narrowed scope decision (not part of primary search). Most recent search date: 2026-07-14 .	—
8	Search — full electronic	Yes	Full PubMed A+B+C v2 FINAL string in	—

	search strategy for at least one database, repeatable.	(content ready)	`docs/protocol/search_string_pubmed_v1.md` (497 records, 2015-01-01 to 2026-06-10) plus the differential-window string (2026-06-11 to 2026-07-14, identical term structure). IEEE and ACM strings in their respective `docs/protocol/search_string_*.md` files. Present at least the PubMed string in full in a supplementary table/box; the others can go in supplementary material.	
9	Selection of sources of evidence — process for selecting sources (screening and eligibility).	Yes (content ready)	Title/abstract screening: Rayyan, single reviewer, sensitivity-favoring 3-gate criteria (`docs/protocol/screening_criteria.md`), TA-E1–E5 exclusion taxonomy. Full-text screening: F1-F5 criteria + E1-E6 exclusion taxonomy (`docs/protocol/screening_fulltext_criteria.md` v1.1). Sole-reviewer status at both T/A and full-text stages, faculty-confirmed accommodation given supervisor availability constraints (`memos/decision_log.md`, "Faculty adjudication delegated" and 2026-07-10 entries) — state this plainly as a disclosed deviation from the two-independent-reviewer standard.	—
10	Data charting process — methods used to chart data (forms, independent/duplicate charting, process for confirming data with investigators).	Yes	Methods, "Data extraction" (`manuscript_v1.md`); full column definitions in Supplementary Material S3. Charting was completed for all 7 included studies (`data/extraction/schema_v1.csv`). Sole reviewer, no independent duplicate charting — disclosed explicitly in Methods, "Limitations acknowledged in the design," as a departure from dual-reviewer convention rather than implied as duplicate charting.	—
11	Data items — list and define all variables sought, with assumptions/simplifications.	Yes	Full column definitions in Supplementary Material S3: clinical domain, XAI method/scope, study design/participants, evaluation type (six-level taxonomy), workflow realism, 4-dimension ecological validity, clinician involvement (evaluation vs. design, kept separate by design), outcomes claimed vs. demonstrated, trust-claim type (calibration vs. plausibility distinction), 5-dimension quality/risk-of-bias, method-justification type (RQ1),	—

			interpretability-framework classification (Doshi-Velez & Kim; Vilone & Longo), 4 regulatory-evidence criteria (RQ3), semantic tags.	
1 2	Critical appraisal of individual sources of evidence — rationale, methods, and how used in synthesis (if done).	Yes (content ready)	Custom 5-dimension rubric (QR1-QR5: Participant Appropriateness, Task Fidelity, Outcome Measurement, Explanation Faithfulness, Reporting Completeness — `data/coding/quality_rubric.md`), adapted from QUADAS-2/RoB 2/TRIPOD elements plus two dimensions with no standard-tool equivalent (rationale: `memos/decision_log.md`, 2026-05-26 entry). Used as a supplementary/exploratory characterization of the corpus, not as an inclusion/exclusion filter — state this explicitly to avoid the appraisal reading as a second screening pass.	—
1 3	Synthesis of results — methods for handling/summarizing charted data.	Yes (content ready)	Narrative synthesis organized by RQ1-RQ3 (JBI methodology, Peters et al. 2020; PRISMA-ScR, Tricco et al. 2018) — not confirmatory hypothesis testing. The originally-planned H1-H6 hypotheses and quality-rubric scores are repositioned as supplementary/exploratory analyses feeding the RQ2 narrative, not primary confirmatory tests (2026-06-10 pivot decision).	—

Results

#	Item	Reported ?	Location / Notes	Page #
1 4	Selection of sources of evidence — numbers screened/assessed/included, reasons for exclusion at each stage, ideally a flow diagram.	Yes	Methods, "Study selection" (`manuscript_v1.md`): 659 identified (497 PubMed + 161 IEEE + 1 ACM) → 650 after deduplication → 650 screened at T/A → 64 forwarded to full-text → 55 assessed at full-text (9 not retrievable, disclosed separately from exclusions) → 48 excluded (E1=22, E2=5, E3=13, E4=8, E5=0, E6=0) → 7 included. The 2026-07-14 post-hoc differential-search supplement (46 additional candidates screened, 0 added) is reported alongside. Flow diagram built	—

			(`docs/figures/prisma_scr_flow_diagram.html`); convert to a static figure image at final typesetting.	
1 5	Characteristics of sources of evidence — present charted characteristics per source, with citations.	Yes	Per-study characteristics charted in Tables 1-3 (`data/synthesis/results_tables_v1.md`) and full per-paper detail in `data/extraction/schema_v1.csv` / `data/extraction/per_paper/`; citations in `references_included_studies.md` (7 entries, DOI-verified).	—
1 6	Critical appraisal within sources of evidence — present appraisal data (if done).	Yes	QR1-QR5 scores charted per included paper in `data/extraction/schema_v1.csv`; referenced narratively in Results (e.g., Xie et al., 2021 scoring maximum on all five dimensions). Used as a supplementary/exploratory characterization, not an inclusion/exclusion filter (Methods, Data extraction).	—
1 7	Results of individual sources of evidence — relevant charted data per included source.	Yes	Per-paper extracted data (XAI method, eval type, realism level, trust claim, etc.) charted in `data/extraction/schema_v1.csv`; charted results presented in Tables 1-3 and synthesized narratively in Results (`manuscript_v1.md`).	—
1 8	Synthesis of results — summarize/present charting results as they relate to the review questions.	Yes	Results (`manuscript_v1.md`): "Methods and Justification (RQ1)," "Evaluation Rigor (RQ2)," "Regulatory Readiness (RQ3)" subsections, each a narrative synthesis of the charted data.	—

Discussion

#	Item	Reported?	Location / Notes	Page #
19	Summary of evidence — main results (overview of concepts/themes/evidence types), link to questions/objectives, relevance to key groups.	Yes	Discussion (`manuscript_v1.md`), opening paragraph: summarizes the corpus-wide pattern (explanations built and computationally measured, but rarely evaluated with any human), links explicitly to RQ1-RQ3, and states relevance to the clinicians and	—

			regulatory audiences this review's findings address.	
20	Limitations — discuss limitations of the scoping review process.	Yes	Methods, "Limitations acknowledged in the design," and Discussion, "What this corpus is small enough, and specific enough, to say" (`manuscript_v1.md`): Embase dropped (no institutional access); CINAHL shelved; English-language-only restriction; single reviewer with no independent duplicate screening/charting; 10 records never retrieved (9 main- corpus + 1 differential- supplement); small final corpus (n=7) framed as a substantive finding rather than an apology.	—
21	Conclusions — general interpretation relative to questions/objectives, implications, next steps.	Yes	Conclusion (`manuscript_v1.md`): interprets findings against RQ1-RQ3, states the review's contribution (governance- compliance standard for EM XAI evaluation), and closes on next steps (clinician-in-the-loop evaluation work this literature currently lacks).	—

Funding

#	Item	Reported?	Location / Notes	Page #
22	Funding — sources of funding for included studies and for the review itself; role of funders.	Yes	No funding was received for this review. State this plainly in the manuscript's Funding/Declarations section (e.g., "This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.").	—

Completion Status Summary

All 22 items are reported as of 2026-07-26. Remaining work before submission is typesetting-dependent, not content-dependent:

- Fill in actual `Page #` values once the manuscript is typeset for a specific journal template.
- Convert the PRISMA-ScR flow diagram (`docs/figures/prisma\_scr\_flow\_diagram.html`, Item 14) to a static figure image in the format the target journal requires.
- Verify item wording and section grouping against the official Tricco et al. (2018) checklist PDF before submission, per this document's standing verification note above.